



7336 - Commission a new R~2500 `IFU` on JWST: Calibrating second-order spectra of NIRCam/WFSS through Hubble Ultra Deep Field

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCam/WFSS F277W Observation of HUDF				
	1	Grism R	NIRCam Wide Field Slitless Spectroscopy	(1) HUDF-WFSS-Center
	2	Grism C	NIRCam Wide Field Slitless Spectroscopy	(1) HUDF-WFSS-Center

ABSTRACT

All NIRCam WFSS scientific results so far are based on first-order spectra, mostly obtained at 3-5 micron. However, NIRCam can also obtain second-order spectroscopy at 2.4-3.0 micron, reaching a medium-high spectral resolution of $R \sim 2500$. Compared with first-order, second-order spectroscopy takes advantage of lower sky background (because of twice higher R) and comparable throughput ($\sim 30\%$). This could offer highly efficient emission-line spectroscopy over $\sim 3\text{-arcmin}^2$ field of FoV, extremely powerful for spatially resolving galaxy kinematics out to $z \sim 5$ as a "poor-man's IFU".

Archival data are insufficient for the calibration of NIRCam WFSS second-order spectroscopy. Only one wavelength calibrator has been observed with very limited dither positions, leaving enormous extrapolation error across the whole FoV.

We propose to fully calibrate the second-order slitless spectroscopy function of NIRCam WFSS. This will be achieved by observing the Hubble Ultra Deep Field (HUDF) through the F277W filter and all four module x grism combinations. HUDF has the highest surface density of accurate spectroscopic redshifts thanks to decades of multi-wavelength observations. We will detect second-order emission lines from galaxies at $z \sim 0-5$ with known redshifts and wavelengths, providing accurate wavelength calibration and sensitivity characterization of second-order spectroscopy to the broader JWST user community. Scientifically, F277W grism is only missing piece for JWST near-IR spectroscopic campaigns in HUDF over Cycle 1-3. The request observation will enhance the JWST legacy in HUDF, opening a new window for wide-field galaxy kinematics survey around and beyond cosmic noon.

OBSERVING DESCRIPTION

We will obtain NIRCam wide-field slitless spectroscopic (WFSS) observations of the Hubble Ultra Deep Field with F277W as the long-wavelength (LW) filter. We will use grism R in obsnum 1, and grism C in obsnum 2.

The primary technical goal of the proposal is to calibrate the 2nd-order spectra of NIRCam/WFSS through F277W filter and all four available grisms (module A / B x grism R / C). The targeted field (Hubble Ultra Deep Field; HUDF) is known for the richness of deep imaging and spectroscopic coverage through multiple ground-based and space telescopes at all wavelength. Specifically, deep HST and JWST imaging data with nearly all major instruments and filters have been obtained for this field, and thousands of accurate spectroscopic redshifts have been obtained through ground-based multiple-object spectroscopy, integral field spectroscopy (notably VLT/MUSE, the MUSE-HUDF survey; Bacon et al. 2017, 2023), JWST NIRSpec observations (primarily through JADES; Eisenstein et al. 2023, Bunker et al. 2023), NIRISS/WFSS observations (through NGDeep survey; Pirzkal et al. 2024), and NIRCam/WFSS observations (through FRESCO using F444W filter, Oesch et al. 2023; GTO-4549 using F356W filter, Rieke et al.).

This program will obtain NIRCam WFSS observation using F277W filter, which is the only missing piece for complete JWST near-infrared spectroscopic investigations in the iconic Hubble Ultra Deep Field.

We will employ 2-integration x 7-group MEDIUM8 readout for all grism exposures, and the exposure time will be 1471 s per dither position. This is to maximize the depth and number of emission-line detections at each dither position without the concern that the data will be overwhelmed by cosmic rays (typically when $t_{\text{exp}} > 1000$ s per each integration). We request (1) two rows of mosaics with Grism R, covering the central 2×2 arcmin² of HUDF for full F277W 1st-order spectroscopy, and (2) two columns of mosaics with Grism C, covering two (nearly) complete 2×2 arcmin² regions along the NE / SW edges of HUDF. We request 4 INTRAMODULEX dithers with each pointing to map out the 2nd-order wavelength calibration functions across the full NIRCam FoV, and ensure the quality and fidelity of co-added spectroscopic data products.

For better calibration purpose, we will obtain F200W imaging observation at the same time as the F277W grism observation. This is because F200W is one of the most commonly used filters at the same time of grism observations (e.g., used by Cycle-1 programs ASPIRE, EIGER, IceAge and Cycle-3 programs COSMOS-3D and NEXUS). The use of same F200W filter will (1) remove the concern of filter-introduced offset for wavelength calibration, and (2) offer time-domain value for the studies of supernovae and AGN in the HUDF region.

With the relatively shallow direct imaging observations after the grism exposures, we plan to use F150W and F250M filter in the short / long-wavelength channel, respectively. F150W data will also offer time-domain value, characterizing the colors of detected transients. F250M imaging data will be useful for the selection of galaxies that could yield emission-line detections at 2.4-2.6 micron, where the grism throughput peaks for 2nd-order spectra. The field has deep NIRCam imaging data at 1-5 micron through JADES including F277W filter, allowing us to detect any galaxy for

which we expect a 5-sigma emission-line detection with the grism exposure.

The observations will be obtained at similar V3 PA of those with GTO-1180, NIRCcam imaging observation of HUDF.

Proposal 7336 - Targets - Commission a new R~2500 `IFU` on JWST: Calibrating second-order spectra of NIRCam/WFSS through H...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	HUDF-WFSS-Center	RA: 03 32 38.3445 (53.1597687d) Dec: -27 46 59.59 (-27.78322d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	
Comments: This is HUDF center Category=Unidentified Description=[Blank field] Extended=NO					

Proposal 7336 - Observation 1 - Commission a new R~2500 `IFU` on JWST: Calibrating second-order spectra of NIRCam/WFSS thro...

Observation	Proposal 7336, Observation 1: Grism R											Tue Mar 11 20:10:11 GMT 2025	
	Diagnostic Status: Warning												
	Observing Template: NIRCam Wide Field Slitless Spectroscopy												
Diagnostics	(Grism R (Obs 1)) Warning (Form): For Module=ALL the default target location is in the gap between the modules.												
	(Grism R (Obs 1)) Warning (Form): Use of only one of GRISMR or GRISMC may result in spectral overlap from multiple sources that can't be corrected. Users should address this issue in their proposal text.												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	HUDF-WFSS-Center	RA: 03 32 38.3445 (53.1597687d) Dec: -27 46 59.59 (-27.78322d) Equinox: J2000				Proper Motion RA: 0 Proper Motion Dec: 0						
	Comments: This is HUDF center Category=Unidentified Description=[Blank field] Extended=NO												
Template	Module		Subarray					Grism (Long Wavelength)					
	ALL		FULL					GRISMR					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	1	2	0.0		88.0		0.0		0.0		DEFAULT		
Dithers	#	Primary Dither Type				Primary Dithers				Subpixel Positions			
	1	INTRAMODULEX				4				NONE			
Direct Image	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	Grism (Long Wavelength)	Exposure Type	Total Dithers	
	1	F150W	F250M	SHALLOW4	6	1	1	311.366		GRISMR	Direct Image	1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	Grism (Long Wavelength)	Exposure Type	Total Dithers	
	1	F200W	F277W	MEDIUM8	7	2	8	5883.75		GRISMR	Grism (Long Wavelength)	4	
	2	F150W	F250M	SHALLOW4	6	1	2	622.733			Out of Field	2	

Proposal 7336 - Observation 1 - Commission a new R~2500 `IFU` on JWST: Calibrating second-order spectra of NIRCam/WFSS thro...

Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 295 to 325 Degrees (V3 295.0 to 325.0) Visits Same PA Background Limited. Background no more than 30th percentile above minimum Aperture PA Offset 2 from 1 by 0 to 5 Degrees (Same offsets in V3)
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Proposal 7336 - Observation 2 - Commission a new R~2500 `IFU` on JWST: Calibrating second-order spectra of NIRCam/WFSS thro...

Observation	Proposal 7336, Observation 2: Grism C										Tue Mar 11 20:10:11 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Wide Field Slitless Spectroscopy											
Diagnostics	(Grism C (Obs 2)) Warning (Form): For Module=ALL the default target location is in the gap between the modules.											
	(Grism C (Obs 2)) Warning (Form): Use of only one of GRISMR or GRISMC may result in spectral overlap from multiple sources that can't be corrected. Users should address this issue in their proposal text.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	HUDF-WFSS-Center		RA: 03 32 38.3445 (53.1597687d) Dec: -27 46 59.59 (-27.78322d) Equinox: J2000			Proper Motion RA: 0 Proper Motion Dec: 0					
	Comments: This is HUDF center Category=Unidentified Description=[Blank field] Extended=NO											
Template	Module		Subarray					Grism (Long Wavelength)				
	ALL		FULL					GRISMC				
Mosaic	Rows	Columns		Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	2	1		58.0		10.0		0.0		0.0		DEFAULT
Dithers	#	Primary Dither Type					Primary Dithers			Subpixel Positions		
	1	INTRAMODULEX					4			NONE		
Direct Image	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
	1	F150W	F250M	SHALLOW4	6	1	1	311.366		GRISMC	Direct Image	1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
	1	F200W	F277W	MEDIUM8	7	2	8	5883.75		GRISMC	Grism (Long Wavelength)	4
	2	F150W	F250M	SHALLOW4	6	1	2	622.733			Out of Field	2

Proposal 7336 - Observation 2 - Commission a new R~2500 `IFU` on JWST: Calibrating second-order spectra of NIRCam/WFSS thro...

Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 295.0 to 325.0 Degrees (V3 295.0 to 325.0) Visits Same PA Offset 0.0 arcsec, 50.0 arcsec Background Limited. Background no more than 30th percentile above minimum Aperture PA Offset 2 from 1 by 0 to 5 Degrees (Same offsets in V3)
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